

Taxes	Let T denote the set of taxes.
Universe	Let U denote the set of economic operations that are not prohibited in a given economic context.
Taxable events	Let $H \subseteq U = \{u \in U \mid u \rightarrow \exists t \text{ such that } \Delta t \neq 0\}$, with $\forall t \in T \exists h \in H \mid t = f(h)$.
Economic operations	Let $O \subseteq U = \{u \in U \mid u \rightarrow \exists h \text{ such that } h = f(u)\}$.
Text	Let τ denote the universal set of texts.
Legal norms	Let $N \subseteq \tau = \{\tau \in \tau \mid \tau \text{ is a legal article}\}$.
Effect on taxable events	Let $A \subseteq N = \{n \in N \mid \exists h \text{ such that } h = f(n)\}$.
Effect on economic operations	Let $\Phi \subseteq N = \{n \in N \mid \exists o \text{ such that } o = f(n)\}$.
Relevance	Let $R = A \cup O$.
Interpretation	Let I_i be an operation $\tau \rightarrow \tau$ such that $I = \{(\tau_1, \tau_2) \in \tau \times \tau \mid \tau_2 \text{ is a common-sense interpretation of } \tau_1\}$.
Interpretable	An element r is interpretable if it contains a fragment $f \in \tau$ such that $\exists \tau_2 \subseteq \tau$ with $f I_f \tau_2$ and $\exists \tilde{\tau}_2 \neq \hat{\tau}_2 \in \tau_2$ where $f(\tilde{\tau}_2) \neq f(\hat{\tau}_2)$.
Discretionary operations	Let $\hat{D} \subseteq U = \{u_1, \dots, u_n\}$ such that the u_i are mutually exclusive and any of them may be executed by an individual.
Discretionary	An element r is discretionary if $\exists \hat{d} \in \hat{D}$ such that $t(h, d_i) \neq t(h, d_j)$ for $i \neq j$.
Differentiator	Let $f : \tau \rightarrow \tau$ such that $(\tau_f, \tau_2) \in \tau \times \tau$ and τ_2 is a hyponym of τ_f .
Differential regime	An element r establishes a differential regime if $\exists f_1 \neq f_2$ such that $t(h, f_1) \neq t(h, f_2)$.

Table 1. Formal Framework for Gray Area Indicators

0.1 Annotation Protocol

Objective The objective is to classify legal articles from Colombian Free Trade Zone regulations according to their potential to generate *gray areas*, as captured by the indicators defined in the formal framework.

Corpus Definition

- **Source:** Laws governing Colombian Free Trade Zones
- **Time period:** 1958–2005
- **Unit of analysis:** Articles
- **Size:** 169 articles

- **Annotators:** 1
- **Output format:** Excel
- **Decision rule:** In cases of doubt, prioritize inclusion (i.e., assign a positive label). The protocol favors exhaustiveness over omission.

0.2 Annotation Scheme

Completeness (Binary)

- **Question:** Does the article contain a fragment that specifies legal or tax treatment for some cases while omitting other relevant situations?
- **Answers:** Yes / No
- **Risk:** When a list of elements (e.g., cases subject to a tax or exemption) is specified but not exhaustive, unlisted yet relevant cases may fall into gray areas and be used to obtain unintended tax treatment.
- **Literature concepts:** Open-ended lists; legal gaps; fragmentation; silence of the law.
- **Example:**
Law 109 of 1985, Article 7. Within the area designated as commercial free trade zones, the development of activities other than those provided for in this Chapter shall not be permitted, nor shall the establishment of private residences or the conduct of retail trade, except in the case of restaurants, cafeterias, and, in general, establishments intended to provide services to people working within the jurisdiction of the respective zone.

Differential Regime (Binary)

- **Question:** Does the article contain a fragment that assigns different tax treatments within the same system based on specific attributes or classifications?
- **Answers:** Yes / No
- **Risk:** When multiple tax rates apply to otherwise comparable entities within the same system, conditional on specific requirements, there is a risk that entities will engage in strategic behavior or artificial arrangements to qualify for the more favorable regime.
- **Literature concepts:** Differential treatment; special tax regimes; exceptions; classification-based distinctions.
- **Example:**
Law 1004 of 2005, Article 5. Tax rate for Free Trade Zone users. As of January 1, 2007, a single income tax rate of fifteen percent (15%) is established on the taxable income of legal entities that are users of a Free Trade Zone.
Paragraph. The income tax rate applicable to commercial users of a Free Trade Zone shall be the prevailing general rate.

Discretion (Binary)

- **Question:** Does the article contain a fragment where the application or enforcement of tax rules depends on administrative judgment rather than explicit legal criteria?
- **Answers:** Yes / No
- **Risk:** When the determination or amount of tax depends on the judgment of an authority or individual, there is scope for discretion that may enable corruption, strategic manipulation, or inducement of error in the evaluation process.
- **Literature concepts:** Administrative discretion; discretion of fiscal authorities; evaluative concepts; legal standards.
- **Example:**
Law 47 of 1981, Article 47. Companies (...) shall export at least sixty percent (60%) of their production in each fiscal year, unless they demonstrate before the Board of Directors of the corresponding zone the need to supply shortages in the domestic market, the occurrence of force majeure or unforeseeable events, circumstances in the external market that have affected demand, or other justified causes that prevent compliance with the export quota.

Interpretation (Multiclass)

- **Question:** Does the article contain a fragment whose wording admits multiple plausible interpretations in common or technical usage, leading to distinct legal or tax outcomes?
- **Answers:** Low / Medium / High
- **Risk:** When the tax outcome depends on the interpretation of terms or provisions, ambiguity can be exploited through selective interpretation, artificial technicalization, or increased complexity to minimize tax liabilities.
- **Literature concepts:** Ambiguity; vagueness; vague concepts; open-textured terms; interpretive uncertainty.
- **Example:**
Law 47 of 1981, Article 47. "...the need to supply shortages in the domestic market, the occurrence of force majeure or unforeseeable events, circumstances in the external market that have affected the demand for the respective product, or other justified causes..."

Relevance (Binary)

- **Question:** Does the article contain a fragment that directly affects tax liabilities or regulates economic operations with a direct impact on tax calculation or payment?

- **Answers:** Yes / No
- **Risk:** Although not a risk per se, economically relevant operations—those that affect tax liabilities—constitute the primary domain in which strategies to reduce tax burdens may arise; where such relevance is absent, the scope for tax planning is significantly reduced.
- **Literature concepts:** Materiality; tax significance; economic relevance.
- **Example:**
Law 105 of 1958, Article 36. Entry into the Zone shall not be permitted for:
a) Explosive or flammable materials; b) Weapons in general; and c) Items specified in the Zone’s regulations, unless expressly authorized by the National Government.